

Question 11a

Create an array of numbers: `const nums = [10, 20, 30]`; Modify the last value in the array and change it to 99.

Solution:

```
const nums = [10, 20, 30];

nums[2] = 99;

console.log(nums);
```

Question 11b

Create a function `getLastValue(array)` that takes an array and returns the last value in the array.

Solution:

```
function getLastValue(arr) {
  return arr[arr.length - 1];
}

let lastArr = [4, 5, 6, 7];
console.log(getLastValue(lastArr));
```

Question 11c

Create a function `arraySwap(array)` that takes an array and returns an array where the first and last values are swapped.

Solution:

```
function arraySwap(arr) {
  const firstValue = arr[0];
  const lastValue = arr[arr.length - 1];

  arr[0] = lastValue;
  arr[arr.length - 1] = firstValue;

  return arr;
}

let egArray = [5, 6, 7, 8];
console.log(arraySwap(egArray));
```

Question 11d

Create a for loop that counts up from 0 to 10, counting by 2.

Solution:

```
for (let i = 0; i <= 10; i += 2) {
  console.log(i);
}
```

Question 11e

Create a for loop that counts down from 5 to 0.

Solution:

```
for (let i = 5; i >= 0; i--) {  
  console.log(i);  
}
```

Question 11f

Redo exercises 11d and 11e using while loops.

Solution:

```
let i = 0;  
while (i <= 10) {  
  console.log(i);  
  i += 2;  
}
```

```
let i = 5;  
while (i >= 0) {  
  console.log(i);  
  i -= 1;  
}
```

Question 11g

Create a loop that takes an array of numbers and creates a new array where each number is increased by 1.

Solution:

```
let arr = [1, 2, 3, 4, 5];  
let arr2 = [];  
  
for (let i = 0; i < arr.length; i++) {  
  arr2.push(arr[i] + 1);  
}  
  
console.log(arr2);
```

Question 11h

Create a function addOne(array) that takes an array of numbers and returns a new array where each number is increased by 1.

Solution:

```
function addOne(arr) {  
  let newArr = [];  
  
  for (let i = 0; i <= arr.length - 1; i++) {  
    newArr.push(arr[i] + 1);  
  }  
  
  return newArr;  
}  
  
const lol = [10, 20, 30, 40];  
console.log(addOne(lol));
```

Question 11i

Create a function `addNum(array, num)` that takes an array of numbers and returns a new array where each number is increased by `num`.

Solution:

```
function addNum(array, num) {
  let arr3 = [];

  for (let i = 0; i <= array.length - 1; i++) {
    arr3.push(array[i] + num);
  }

  return arr3;
}

const arr = [1, 2, 3, 4, 5];
console.log(addNum(arr, 5));
```

Question 11j

Create a function `addArrays(array1, array2)` that takes two arrays of numbers and adds each number in the arrays together.

Solution:

```
function addArrays(array1, array2) {
  let arr3 = [];

  for (let i = 0; i <= array1.length - 1; i++) {
    arr3.push(array1[i] + array2[i]);
  }

  return arr3;
}

const arr = [10, 20, 30, 40];
const arr2 = [10, 20, 30, 40];
console.log(addArrays(arr, arr2));
```

Question 11k

Create a function `countPositive(nums)` that takes an array of numbers and returns how many numbers are greater than 0.

Solution:

```
function countPositive(nums) {
  let count = 0;

  for (let i = 0; i < nums.length; i++) {
    if (nums[i] > 0) {
      count++;
    }
  }

  return count;
}

const arr = [0, 10, 20, 30, 40];
console.log(countPositive(arr));
```

Question 11l

Create a function `minMax(nums)` that returns the minimum and maximum values using a loop.

Solution:

```
function minMax(nums) {
  let minNum = nums[0];
  let maxNum = nums[0];

  for (let i = 0; i < nums.length; i++) {
    if (nums[i] < minNum) minNum = nums[i];
    if (nums[i] > maxNum) maxNum = nums[i];
  }

  return {
    minimumNumber: minNum,
    maximumNumber: maxNum
  };
}

const arr = [4, 5, 6, 9, 10];
console.log(minMax(arr));
```

Question 11m

Update `minMax(nums)` to handle an empty array and a single element array.

Solution:

```
function minMax(nums) {
  let minNum = nums[0];
  let maxNum = nums[0];

  if (minNum === undefined) {
    return null;
  }

  for (let i = 0; i < nums.length; i++) {
    if (nums[i] < minNum) minNum = nums[i];
    if (nums[i] > maxNum) maxNum = nums[i];
  }

  return {
    minimumNumber: minNum,
    maximumNumber: maxNum
  };
}

const arr = [];
console.log(minMax(arr));
```

Question 11n

Create a function `countWords(words)` that returns an object with the number of occurrences of each string.

Solution:

```
function countWords(words) {  
  let count = {};  
  
  for (let i = 0; i < words.length; i++) {  
    if (count[words[i]] === undefined) {  
      count[words[i]] = 1;  
    } else {  
      count[words[i]]++;  
    }  
  }  
  
  return count;  
}  
  
const arr = ["food", "workout", "nap", "code", "food", "sleep"];  
console.log(countWords(arr));
```